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CFRP fuselage frame with securement to vertical tail fin

Abstract

A securement assembly for securing a vertical tail fin assembly to an aircraft includes a first lug member secured to the vertical tail fin assembly. The securement assembly further includes a first clevis member. The first clevis member includes a first end portion of the first clevis member is engaged to the first lug member. A second end portion of the first clevis member is secured to a first fuselage frame constructed of a composite material with a first fastener which extends through the second end portion and the first fuselage frame in a first direction transverse to the first fuselage frame.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8348196	12/2012	Campana	N/A	N/A
9776704	12/2016	Rufino	N/A	B29C 70/384
10046848	12/2017	Folch Cortes et al.	N/A	N/A
2010/0155532	12/2009	Ariza Martin et al.	N/A	N/A
2010/0264271	12/2009	Cortes et al.	N/A	N/A
2011/0089292	12/2010	Williams	29/525.01	B64C 1/26
2012/0091275	12/2011	Vera Villares	244/131	B64C 1/26
2015/0034765	12/2014	Gonzalez Gozalbo	244/120	B64C 9/00
2020/0023934	12/2019	Martino-Gonzalez	N/A	B64C 5/02
2020/0156758	12/2019	Colmenarejo Matellano	N/A	B64C 5/06
2021/0371080	12/2020	Campana	N/A	B64C 7/00

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
109720544	12/2018	CN	N/A
2444315	12/2011	EP	N/A
3078586	12/2015	EP	N/A
3584152	12/2018	EP	N/A
3653490	12/2019	EP	N/A
2018-8679	12/2017	JP	N/A

OTHER PUBLICATIONS

https://www.researchgate.net/figure/Airbus-A300-600-vertical-stabilizer-construction_fig1_279693318, Jan. 2006. cited by applicant

Communication Pursuant to Rule 62 EPC for EP Application No. 21169331.2 dated Oct. 14, 2021. cited by applicant

Extended European Search Report for EP Application No. 21183492.4 dated Jan. 18, 2022. cited by applicant

English version of JP 2018-8679. cited by applicant

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/064,711, entitled “CFRP FUSELAGE FRAME WITH SECUREMENT TO VERTICAL TAIL FIN” and filed Aug. 12, 2020, the entire disclosure of which is hereby incorporated by reference herein.

FIELD

(1) This disclosure relates to securement of a vertical fin assembly of an aerospace vehicle to the aerospace vehicle and more particularly, to securement of the vertical fin assembly to a fuselage frame of an aircraft.

BACKGROUND

(2) There is a concern with respect to the occurrence of corrosion of an aluminum fuselage frame with an aluminum frame being attached to a composite fiber reinforced polymer (“CFRP”) skin of an aircraft. In addition, with constructing an aircraft with utilizing, for example, an aluminum frame with a CFRP skin, fatigue issues arise due to the material differences and their different thermal expansion effects. Aluminum, for example, has a thermal expansion co-efficient greater than that of the CFRP. Aluminum experiences contraction in cold conditions and expands in hot conditions relative to the CFRP material which in comparison is thermally neutral. As a result of using different materials for the frame and skin structures, these structures experience compression and tension forces with the aircraft experiencing differing temperatures in operation of the aircraft. Moreover, with respect to production, metallic frames, such as aluminum fuselage frames, require de-burring after drilling to prevent fatigue cracking; require fay surface sealing prior to installation to prevent corrosion; and impose expanded time demands in production.

(3) As a result, there is a need to reduce costs in aircraft production associated with corrosion protection measures being taken with use of different materials such as aluminum for frames and CFRP for skin of the aircraft and to reduce thermal fatigue as a result of thermal expansion effects on interconnected structures in the aircraft which are constructed of different materials each having a different thermal expansion co-efficient. Also, there is a need to shorten production time with respect to the time imparted to production with use of metallic frames.

SUMMARY

(4) An example includes a securement assembly for securing a vertical fin assembly to an aircraft includes a first lug member secured to the vertical fin assembly. The securement assembly includes a first clevis member. A first end portion of the first clevis member is engaged to the first lug member. A second end portion of the first clevis member is secured to a first fuselage frame constructed of a composite material, with a first fastener which extends through the second end portion and the first fuselage frame in a first direction transverse to the first fuselage frame.

(5) An example includes a method for securing a vertical fin assembly to an aircraft, which includes securing a first lug member, which is secured to the vertical fin assembly, to a first end portion of a first clevis member. The method further includes securing a second end portion of the first clevis member to a first fuselage frame constructed of a composite material, with a first fastener which extends through the second end portion of the first clevis member and the first fuselage frame in a first direction transverse to the first fuselage frame.

(6) The features, functions, and advantages that have been discussed can be achieved independently

in various embodiments or may be combined in yet other embodiments further details of which can be seen with reference to the following description and drawings.

Description

BRIEF SUMMARY OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an aircraft;
- (2) FIG. 2 is a perspective partial view, in a direction toward a forward portion of the aircraft of FIG. 1, of composite fuselage frames secured to a portion of a vertical tail assembly of the aircraft, without composite fuselage skin being shown;
- (3) FIG. 3 is a perspective partial view, in a direction toward an aft portion of the aircraft of FIG. 1, of composite fuselage frames secured to a portion of a vertical tail assembly of the aircraft of FIG. 2;
- (4) FIG. 4 is an end elevation partial view, in a direction toward the forward portion of the aircraft of FIG. 1, of the composite fuselage frames secured to a portion of the vertical tail assembly of the aircraft of FIG. 2;
- (5) FIG. 5 is a top partial plan view of two composite fuselage frames of FIG. 2, with each fuselage frame having two clevis members positioned spaced apart from one another along each of the two composite fuselage frames without any of the vertical tail assembly or lug members present;
- (6) FIG. 6 is a cross section view along line 6-6 as seen in FIG. 5 including a lug member secured to a clevis member extending through composite fuselage skin;
- (7) FIG. 7 is a cross section view along line 7-7 as shown in FIG. 5 including a lug member secured to a clevis member;
- (8) FIG. 8 is a schematic view of a fiber arrangement within the composite material of fuselage frame as representatively seen at location of circle identified with see FIG. 8 in FIG. 4;
- (9) FIG. 9 is a schematic cross section view of a fuselage and vertical tail fin assembly of the aircraft of FIG. 1 showing reactive forces experienced by securement assemblies which secure fuselage frames to vertical tail fin assembly of aircraft with the application of a lateral force applied to the vertical tail fin assembly; and
- (10) FIG. 10 is a flow chart of a method for securing a vertical tail fin assembly to an aircraft.

DESCRIPTION

- (11) In referring to FIG. 1, aircraft 10 includes fuselage assembly 12, wing assemblies 14 and vertical tail fin assembly 16. In fabrication of aircraft 10, vertical tail fin assembly 16 includes a support structure (not shown) of spars and ribs to which skin 18 of aircraft 10 secures. Vertical tail fin assembly 16 further secures to fuselage assembly 12 of aircraft 10 with being connected to fuselage frames. In construction of aircraft 10, as discussed earlier, fabricators have utilized metal, such as aluminum, to construct fuselage frames. Metallic fuselage frames were similar to the configurations of fuselage frames, such as for example, first, second, third and fourth fuselage frames, 20a, 20b, 20c and 20d, as seen in FIGS. 2-4. The number of fuselage frames to which vertical tail fin assembly 16 is secured can vary depending on the design of the particular aircraft 10.
- (12) In the present disclosure, first through fourth fuselage frames 20a-20d, are now constructed with CFRP material, instead of metal, which avoids drawbacks, as discussed earlier associated with metallic fuselage frames. With utilizing CRFP for constructing fuselage frames there is no longer a need to take corrosion resistance preventative measures with respect to a metallic fuselage frame. In addition, with a CFRP constructed fuselage frame there is no longer a need for inspections related to material fatigue with respect to the first through fourth fuselage frames 20a-20d and fuselage skin 22, with first through fourth fuselage frames 20a-20d and skin 18 of aircraft 10 both being constructed of similar CFRP materials resulting in first through fourth fuselage frames 20a-

20d and skin **18** having similar thermal expansion co-efficient characteristics. Moreover, extended fabrication times of aircraft **10** are reduced with eliminating forging order delays with respect to fabrication of metallic fuselage frames being removed from the fabrication scheduling.

(13) These drawbacks are overcome with first through fourth fuselage frames **20a-20d**, in this example, and fuselage skin **22** now both being constructed of CFRP material. With the benefits provided with using CFRP material used for fuselage frames such as, **20a-20d**, securement assembly **24**, to be discussed herein, is needed for securing vertical tail fin assembly **16** to aircraft **10** through first through fourth fuselage frames **20a-20d** of fuselage assembly **12**. Securement assembly **24** provides needed transferring of shear loads from vertical tail fin assembly **16** to first through fourth fuselage frames **20a-20d**, of the present example, which are constructed of CFRP material.

(14) Securement of vertical tail fin assembly **16** to fuselage frames constructed of metal, was accomplished with bolting of vertical tail fin assembly **16** in a generally vertical direction relative to vertical tail fin assembly **16** and the metal fuselage frames. However, this configuration of securement, with respect to a fuselage frame now constructed of CFRP material, does not provide optimal shear resistance with respect to shear loadings transmitted from vertical tail fin assembly **16** to, in this example, first through fourth fuselage frames **20a-20d** as a result of aircraft **10** operations. As a result, securement assembly **24**, as seen in FIGS. 2-7, and described herein, provides an optimal securement of vertical tail fin assembly **16** to, in this example, first through fourth fuselage frames **20a-20d** constructed of CFRP for resistance to shear loads originating from vertical tail fin assembly **16**.

(15) In the present example, of securing vertical tail fin assembly **16** to four, first through fourth, fuselage frames **20a-20d** herein, each of first through fourth fuselage frames **20a-20d** has a pair of securement assemblies **24** spaced apart on each of first through fourth fuselage frames **20a-20d** securing vertical tail fin assembly **16** to each of first through fourth fuselage frames **20a-20d**. In referring to FIG. 2 a starboard side **26** of aircraft **10** perspective view of first through fourth fuselage frames **20a-20d** is seen and in FIG. 3 a port side **28** of aircraft **10** perspective view of first through fourth fuselage frames **20a-20d** is seen. The two views provide a view of each pair of securement assemblies **24** with respect to each of first through fourth fuselage frames **20a-20d**. First pair **30** of securement assemblies **24** are positioned on first fuselage frame **20a**, second pair **32** of securement assemblies **24** are positioned on second fuselage frame **20b**, third pair **34** of securement assemblies **24** are positioned on third fuselage frame **20c** and fourth pair **36** of securement assemblies **24** are positioned on fourth fuselage frame **20d**.

(16) In the present example, first and fourth pairs **30** and **36** of securement assemblies **24** are configured the same and are optimal in resisting shear forces transmitted from vertical tail fin assembly **16** that have resulting shear loads in forward direction **38** and aft direction **40**, such as seen in FIGS. 2 and 3. In contrast, in this example, second and third pairs **32** and **34** of securement assemblies **24** are configured the same and are optimal in resisting shear forces transmitted from vertical tail fin assembly **16** that have resulting shear loads oriented in starboard direction **42** or in port direction **44**. The difference between the first pair **30** and fourth pair **36** of securement assemblies **24**, on the one hand, and second pair **32** and third pair **34** of securement assemblies **24**, on the other hand, is the orientation of a pin used in the securement assemblies **24** to lug members to be discussed. In first pair **30** and fourth pair **36** of securement assemblies **24** the pin, for each securement assembly **24** is positioned to extend along first fuselage frame **20a** and fourth fuselage frame **20d**, respectively. In second pair **32** and third pair **34** of securement assemblies **24** the pin, for each securement assembly **24** is positioned to extend transverse to second fuselage frame **20b** and third fuselage frame **20c**, respectively.

(17) In the present example, first pair **30** of securement assemblies **24** is positioned on first fuselage frame **20a** and second pair **32** of securement assemblies **24** is positioned on second fuselage frame **20b** as seen in FIGS. 2 and 3. It should be appreciated the location of first fuselage frame **20a**

carrying first pair **30** of securement assemblies **24** is not restricted to being positioned as the most aft position **48** of first through fourth fuselage frames **20a-20d** which secure to vertical tail fin assembly **16** and the location of fourth fuselage frame **20d** carrying fourth pair **36** of securement assemblies **24** is not restricted to being positioned on the most forward position **50** of first through fourth fuselage frames **20a-20d** which secure to vertical tail fin assembly **16**. Similarly, second pair **32** of securement assemblies **24** is not restricted to being positioned on second fuselage frame **20b** positioned between first and fourth pairs **30, 36** of securement assemblies **24**, which are positioned on first and fourth fuselage frames **20a** and **20d**, respectively. Similarly, third pair **34** of securement assemblies **24** is not restricted to being positioned on third fuselage frame **20c** positioned between the first and fourth pairs **30, 36** securement assemblies, which are positioned on first and fourth fuselage frames **20a** and **20d**, respectively. In the present example described herein, first and fourth pairs **30, 36** of securement assemblies **24** are positioned on first and fourth fuselage frames **20a** and **20d** which are positioned most aft position **48** and most forward position **50** of aircraft **10**, respectively, for securing to vertical tail fin assembly **16**. Second and third pairs **32, 34** are positioned on second and third fuselage frames **20b** and **20c**, respectively, which are positioned between first and fourth fuselage frames **20a** and **20d**, for securing vertical tail fin assembly **16**.

(18) In referring to FIGS. 2-7, securement assembly **24** for securing vertical tail fin assembly **16** to aircraft **10** by way of securing to, in this example, first through fourth fuselage frames **20a-20d**, includes first lug member **46** is secured (not shown) to vertical tail fin assembly **16**, such as for example bolting (not shown) first lug member **46** to a framework of spars and ribs (not shown) of vertical tail fin assembly **16**. First clevis member **52**, positioned on first fuselage frame **20a**, has a first end portion **54** of first clevis member **52** engaged to first lug member **46**, as seen in FIG. 6 and which will be discussed in further detail. First clevis member **52** further includes second end portion **56** of the first clevis member **52** which is secured to first fuselage frame **20a**, which is constructed of composite material, CFRP. First fastener **58** extends through second end portion **56**, as will be further discussed, and first fuselage frame **20a** in first direction **60** transverse to first fuselage frame **20a**. In this example, first fastener **58** includes a bolt and nut assembly.

(19) As seen in FIG. 6, first clevis member **52** has first prong **62**, which defines first opening **64** at first end portion **54** of first clevis member **52** and has second prong **66** which defines second opening **68** at first end portion **54** of first clevis member **52**. First lug member **46** is positioned between first prong **62** and second prong **66** of first clevis member **52**. First pin **70** extends through first opening **64** of first prong **62**, second opening **68** of second prong **66** and through first lug opening **72** defined by and through first lug member **46** engaging first clevis member **52** to first lug member **46** such that first pin **70** extends in first direction **60** transverse to first fuselage frame **20a**. Second end portion **56** of first clevis member **52** includes first securement flange **74** which extends along forward side **76** of first fuselage frame **20a** with first fastener **58** extending through first securement flange **74** and first fuselage frame **20a**. Positioning first fastener **58**, as seen in FIGS. 4 and 6, extending in first direction **60** transverse to first fuselage frame **20a**, first clevis member **52** is secured to first fuselage frame **20a**, which is constructed of CFRP, to optimally secure first securement flange **74** to first fuselage frame **20a** so as to optimally confront shear loadings being received from vertical tail fin assembly **16**.

(20) First pair **35** of clevis members, as seen in FIG. 5, includes first clevis member **52** and second clevis member **78** secured to first fuselage frame **20a** wherein first clevis member **52** and second clevis member **78** are positioned spaced apart from one another in second direction **80** along first fuselage frame **20a**. First clevis member **52** has first prong **62** which defines first opening **64** at first end portion **54** of first clevis member **52** and second clevis member **78** has first prong **82** which defines first opening **84** at first end portion **89**, as seen in FIG. 4, of second clevis member **78** similar to that of first end portion **54** of first clevis member **52**. First clevis member **52** has second prong **66** which defines second opening **68** at first end portion **54** of first clevis member **52** and second clevis member **78** has second prong **86** which defines second opening **88**, as seen in FIG. 5,

at first end portion **89**, as shown in FIG. **4**, of second clevis member **78** similar to that of first end portion **54** of first clevis member **52**, as seen in FIG. **6**.

(21) First lug member **46** is positioned between first prong **62** and second prong **66** of first clevis member **52** of first pair **35** of first clevis member **52** and second clevis member **78**. Second lug member **90**, as seen in FIG. **4** is positioned between first prong **82** and second prong **86** of FIG. **5**, of second clevis member **78** of first pair **35** of first clevis member **52** and second clevis member **78**. First pin **70**, as seen in FIG. **6**, extends through first opening **64** of first prong **62** of first clevis member **52**, second opening **68** of second prong **66** of first clevis member **52** and through first lug opening **72** defined by and through first lug member **46** such that first pin **70** extends in first direction **60** transverse to first fuselage frame **20a**. Second pin (not shown) extends through first opening **84** of first prong **82** of second clevis member **78**, second opening **88** of second prong **86** of second clevis member **78** and through second lug opening (not shown) defined by and through second lug member **90**, as seen in FIG. **4**, such that second pin (not shown) extends in first direction **60** transverse to first fuselage frame **20a**. The configuration of second clevis member **78** engaged to second lug member **90** is similar as shown in FIG. **6** with respect to first clevis member **52** securing to first lug member **46**.

(22) Second end portion **56** of first clevis member **52** includes, as seen in FIG. **6**, first securement flange **74** and second end portion **91** of second clevis member **78** includes second securement flange **92** as seen in FIG. **4**. First securement flange **74** extends along forward side **76** of first fuselage frame **20a** with first fastener **58**, of FIG. **6**, extending through first securement flange **74** and first fuselage frame **20a** in first direction **60** transverse to first fuselage frame **20a**. Second securement flange **92** extends along forward side **76** of first fuselage frame **20a** with second fastener **94**, as seen in FIG. **4**, extending through second securement flange **92** and first fuselage frame **20a** in first direction **60** transverse to first fuselage frame **20a**, similar to the configuration of first securement flange **74** as seen in FIG. **6**.

(23) Third clevis member **96**, as seen in FIGS. **5** and **7**, is secured to, in this example, to second fuselage frame **20b**, constructed of composite material, CFRP, spaced apart from first clevis member **52** along length **L** of aircraft **10**, which has first prong **98** which defines first opening **100** at first end portion **102** of third clevis member **96** and has a second prong **104** which defines second opening **106** at first end portion **102** of third clevis member **96**. Third lug member **108** is positioned between first prong **98** and second prong **104** of third clevis member **96**. Third pin **110** which extends through first opening **100** of first prong **98**, second opening **106** of second prong **104** of third clevis member **96** and through third lug opening (not shown) defined by and through third lug member **108** engaging third clevis member **96** to third lug member **108** such that third pin **110** extends in third direction **112**, as seen in FIG. **5**, along second fuselage frame **20b**. Second end portion **114**, as seen in FIG. **7**, of third clevis member **96** includes third securement flange **116** which extends along forward side **118** of second fuselage frame **20b** with third fastener **120**, such as for example a bolt with a nut, extending through third securement flange **116** and second fuselage frame **20b** in fourth direction **122** transverse to second fuselage frame **20b**.

(24) Further included in this example is second pair of clevis members **37**, as seen in FIG. **5**, including third clevis member **96**, as described above, and fourth clevis member **124** secured to second fuselage frame **20b**, which is also constructed of a composite material, CFRP. Third clevis member **96** and fourth clevis member **124** are positioned spaced apart from one another in third direction **112** along second fuselage frame **20b**. As described above third clevis member **96** has first prong **98** which defines first opening **100** at first end portion **102** of third clevis member **96** and fourth clevis member **124** has first prong **126** which defines first opening **127** at first end portion (not shown) of fourth clevis member **124**. First end portion (not shown) of fourth clevis member **124** is similar to first end portion **102** of third clevis member **96** shown in FIG. **7**. Third clevis member **96**, as discussed earlier and seen in FIG. **5**, has second prong **104** which defines second opening **106** at first end portion **102** of third clevis member **96**. Fourth clevis member **124** has

second prong **128** which defines second opening **130** at first end portion (not shown) of fourth clevis member **124**.

(25) Third lug member **108** is positioned between first prong **98** and second prong **104** of third clevis member **96**. Fourth lug member **132**, as seen in FIG. 3, is positioned between first prong **126** and second prong **128** of fourth clevis member **124** of FIG. 5. Third pin **110**, as seen in FIG. 7, extends through first opening **100** of first prong **98** of third clevis member **96**, second opening **106** of second prong **104** of third clevis member **96** and through third lug opening (not shown) defined by and through third lug member **108**, such that third pin **110** extends in third direction **112**, as shown in FIG. 5, along second fuselage frame **20b**. Fourth pin (not shown) but similar to third pin **110** as seen in FIG. 7, extends through first opening **127** of first prong **126** of fourth clevis member **124**, second opening **130** of second prong **128** of fourth clevis member **124**, of FIG. 5, and through a fourth lug opening (not shown) defined by and through fourth lug member **132**, as seen in FIG. 3, such that fourth pin (not shown) extends in third direction **112** along second fuselage frame **20b** as seen in FIG. 5.

(26) Second end portion **114** of third clevis member **96**, as seen in FIG. 7, includes third securement flange **116** and second end portion (not shown) of fourth clevis member **124** includes fourth securement flange **134**, as seen in FIG. 2. Second end portion (not shown) of fourth clevis member **124** and fourth securement flange **134** are similar to second end portion **114** of third clevis member **96** and third securement flange **116** as shown in FIG. 7. Third securement flange **116** extends along forward side **118**, as seen in FIGS. 2 and 7, of second fuselage frame **20b** with third fastener **120** extending through third securement flange **116** and second fuselage frame **20b** in fourth direction **122** transverse to second fuselage frame **20b**. Fourth securement flange **134**, as seen in FIG. 2, extends along forward side **118** of second fuselage frame **20b** with fourth fastener (not shown) however similar to that of third fastener **120** of FIG. 7 extending through the fourth securement flange **134** and second fuselage frame **20b** in fourth direction **122** transverse to second fuselage frame **20b**, similar to third securement flange **116** securement arrangement as seen in FIG. 7.

(27) First through fourth fuselage frames **20a-20d**, in this example, are constructed with composite material such as CFRP. In the present example, the composite material has a five ply configuration for each layer of composite material used in constructing first through fourth fuselage frames **20a-20d**, wherein in this example multiple layers of composite material are used. In referring to FIG. 8 one ply has a nonlinear fiber configuration **136** of fibers. In constructing each layer of composite material, one ply of composite material contains the nonlinear fiber configuration **136** of fibers which includes twenty percent (20%) of the plies for that layer. In this example, nonlinear fiber configuration **136** of fibers extend in a curved direction having a radial axis R. Two plies of composite material have a first linear fiber configuration of fibers **138** which extend within an angular range which includes plus or minus five degrees of plus thirty degrees of being angularly displaced from radial axis R. The angular displacement is represented as angle "A". Two plies of composite material have first linear fiber configuration of fibers **138** which includes about forty percent (40%) of the plies for that layer. Another two plies of composite material for the layer includes a second linear fiber configuration of fibers **140** which extend within an angular range which includes plus or minus five degrees of minus thirty degrees of being angularly displaced from radial axis R. The angular displacement is represented as angle "B". This example of fiber configuration in the composite material provides a construction that is lighter in weight than a metal counterpart fuselage frame and provides required resistance to shear loads originating from vertical tail fin assembly **16** as a result of aircraft **10** operations.

(28) In referring to FIG. 9, a schematic cross section of aircraft **10** is shown with fuselage assembly **12** and vertical tail fin assembly **16** in viewing toward aft position **48** of aircraft **10**. An example of reactant shear force with respect to first through fourth fuselage frames **20a-20d** during operation of aircraft **10** is shown. In operation of aircraft **10**, an aerodynamic operational force F can be

applied, as in this example, from starboard side **26** of aircraft **10**. The aerodynamic operational force **F** resultant force load is transmitted through vertical tail fin assembly **16** such that fuselage frames such as **20a-20d** provide reactant forces of **F1** in a direction toward vertical tail fin assembly **16** on port side **28** of vertical tail fin assembly **16** of aircraft **10** and of **F2** in a direction away from vertical tail fin assembly **16** on starboard side **26** of vertical tail fin assembly **16** of aircraft **10** in countering aerodynamic operational force **F**. With aerodynamic operational force **F** reversed in direction reactant forces **F1** and **F2** are reversed in direction. In the example of securement assemblies **24** described herein, second and third pairs **32** and **34** of securement assemblies **24** positioned on fuselage frames **20b** and **20c**, respectively, provide optimal load transfers between fuselage frames **20b** and **20c** in addressing the aerodynamic operational force **F** experienced from starboard or port sides **26**, **28** of aircraft **10**. First and fourth pairs **30** and **36** of securement assemblies **24** positioned on fuselage frames **20a** and **20d**, respectively, provide optimal load transfers between fuselage frames **20a** and **20d** and vertical tail fin assembly **16** in addressing longitudinal operational forces on vertical tail fin assembly **16** along length **L** of aircraft **10**.

(29) In referring to FIG. **10**, method **142** for securing vertical tail fin assembly **16** to aircraft **10** includes securing **144** first lug member **46**, which is secured to vertical tail fin assembly **16**, to first end portion **54** of first clevis member **52**. Method **142** further includes securing **146** second end portion **56** of first clevis member **52** to first fuselage frame **20a** constructed of composite material with first fastener **58** which extends through second end portion **56** of first clevis member **52** and first fuselage frame **20a** in first direction **60** transverse to first fuselage frame **20a**.

(30) Securing **144** of first lug member **46** to first end portion **54** of first clevis member **52** includes first clevis member **52** having first prong **62** which defines first opening **64** at first end portion **54** of first clevis member **52** and has second prong **66** which defines second opening **68** at first end portion **54** of first clevis member **52**, as seen in FIG. **5**. First lug member **46** is positioned between first prong **62** and second prong **66** of first clevis member **52** and defines first lug opening **72** through first lug member **46**, as seen in FIG. **6**. First pin **70** extends through first opening **64** of first prong **62**, second opening **68** of second prong **66** and through first lug opening **72** of first lug member **46**, such that first pin **70** extends in first direction **60** transverse to first fuselage frame **20a**.

(31) Method **142** for securing further includes first pair **35** of clevis members including first clevis member **52** and second clevis member **78**, as seen in FIG. **5**, wherein second clevis member **78** is secured to first fuselage frame **20a** spaced apart from first clevis member **52** in second direction **80** along first fuselage frame **20a**. Second pin (not shown) extends through first opening **84** of first prong **82** of first portion of the second clevis member **78**, second opening **88** of second prong **86** of first end portion **89** of second clevis member **78** and through second lug opening (not shown) of second lug member **90**, such that second pin (not shown) extends in first direction **60** transverse to first fuselage frame **20a**. Second fastener **94** extends through second end portion **91**, as seen in FIG. **4**, of second clevis member **78** and through first fuselage frame **20a** in first direction **60** transverse to first fuselage frame **20a** securing second clevis member **78** to first fuselage frame **20a**.

(32) Method **142** further includes securing of third lug member **108**, as seen in FIG. **7**, to first end portion **102** of third clevis member **96** with third securement flange **116** of a second end portion **114** of the third clevis member **96** secured to second fuselage frame **20b**, which is spaced apart from first fuselage frame **20a**. Third clevis member **96** includes first prong **98** which defines first opening **100** at first end portion **102** of third clevis member **96** and has second prong **104** which defines second opening **106** at first end portion **102** of third clevis member **96**, as seen in FIG. **5**. Third lug member **108** is positioned between first prong **98** and second prong **104** of third clevis member **96** and defines third lug opening (not shown) through third lug member **108**. Third pin **110** extends through first opening **100** of first prong **98**, second opening **106** of second prong **104** and through third lug opening (not shown) of third lug member **108**, such that third pin **110** extends in third direction **112** along second fuselage frame **20b**. Third fastener **120** extends through third securement flange **116** of second end portion **114** of third clevis member **96** and second fuselage

frame **20b** in fourth direction **122** transverse to second fuselage frame **20b**, as seen in FIG. 7.

(33) Second pair of clevis members **37** including third clevis member **96** and fourth clevis member **124**, as seen in FIG. 5, wherein fourth clevis member **124** is secured to second fuselage frame **20b** spaced apart from third clevis member **96** in third direction **112** along second fuselage frame **20b**. Fourth clevis member **124** has first prong **126** which defines first opening **127** at first end portion (not shown) of fourth clevis member **124** and has second prong **128** which defines second opening **130** at first end portion (not shown) of fourth clevis member **124**. First end portion (not shown) of fourth clevis member **124** is similar to first end portion **102** of third clevis member **96** shown in FIG. 7. Fourth lug member **132**, as seen in FIG. 3, is positioned between first prong **126** and second prong **128** of fourth clevis member **124** and defines fourth lug opening (not shown) through fourth lug member **132**. Fourth pin (not shown), similar to third pin **110** as seen in FIG. 7, extends through first opening **127** of first prong **126**, second opening **130** of second prong **128** and through fourth lug opening (not shown) of the fourth lug member **132**, such that the fourth pin (not shown) extends in third direction **112** along second fuselage frame **20b**. Fourth fastener (not shown), similar to that of third fastener **120** of FIG. 7, extends through fourth securement flange **134**, as seen in FIG. 2, of second end portion (not shown) of fourth clevis member **124** and second fuselage frame **20b** in fourth direction **122** transverse to second fuselage frame **20b**, similar to third securement flange **116** securement arrangement as seen in FIG. 7.

(34) The composite material used in construction of first through fourth fuselage frames **20a-20d** include five ply configurations for a layer, wherein multiple layers are employed in this example, as described earlier, which includes one ply having a nonlinear fiber configuration **136** of fibers, as seen in FIG. 8. Two plies have a first linear fiber configuration of fibers **138** which extends within an angular range which includes plus or minus five degrees of plus thirty degrees of angular displacement from radial axis R of nonlinear fiber configuration **136** of fibers. Another two plies have a second linear fiber configuration of fibers **140** which extends within an angular range which includes plus or minus five degrees of minus thirty degrees of angular displacement from radial axis R of nonlinear fiber configuration **136** of fibers.

(35) While various embodiments have been described above, this disclosure is not intended to be limited thereto. Variations can be made to the disclosed embodiments that are still within the scope of the appended claims.

Claims

1. A securement assembly for securing a vertical tail fin assembly to an aircraft, comprising: a first lug member secured to the vertical tail fin assembly; and a first clevis member, wherein: a first end portion of the first clevis member is engaged to the first lug member; and a second end portion of the first clevis member is secured to a first fuselage frame, wherein the first fuselage frame is constructed of a composite material, wherein the second end portion of the first clevis member overlaps part of the first fuselage frame in which a first fastener extends through the second end portion of the first clevis member and the first fuselage frame in a first direction transverse to the first fuselage frame, wherein the composite material comprises a five-ply configuration which includes: one ply having a nonlinear fiber configuration; two plies having a first linear fiber configuration which extends within an angular range which includes plus or minus five degrees of plus thirty degrees of angular displacement from a radial axis of the nonlinear fiber configuration; and another two plies having a second linear fiber configuration which extends within an angular range which includes plus or minus five degrees of minus thirty degrees of angular displacement from the radial axis of the nonlinear fiber configuration.
2. The securement assembly of claim 1, wherein the first end portion of the first clevis member defines a first opening along a first clevis axis in which a first pin extends through the first opening of the first clevis member and the first lug member to secure together the first clevis member and

the first lug member, the securement assembly further comprising: a second lug member secured to the vertical tail fin assembly; and a second clevis member, wherein: a first end portion of the second clevis member is engaged to the second lug member, wherein the first end portion of the second clevis member defines a second opening along a second clevis axis in which a second pin extends through the second opening of the second clevis member and the second lug member to secure together the second clevis member and the second lug member, wherein the first pin and the second pin are spaced apart; and a second end portion of the second clevis member is secured to a second fuselage frame, wherein the second fuselage frame is constructed of the composite material, wherein the second end portion of the second clevis member overlaps part of the second fuselage frame in which a second fastener extends through the second end portion of the second clevis member and the second fuselage frame in the first direction transverse to the second fuselage frame.

3. The securement assembly of claim 2, wherein: the first opening of the first clevis member is further defined as a plurality of first openings; the first clevis member has a first prong which defines one of the plurality of first openings at the first end portion of the first clevis member and has a second prong which defines another one of the plurality of first openings at the first end portion of the first clevis member; and the first lug member is positioned between the first prong and the second prong of the first clevis member.

4. The securement assembly of claim 3, wherein the first pin extends through the first opening of the first prong, the first opening of the second prong and through a first lug opening defined by and through the first lug member engaging the first clevis member to the first lug member such that the first pin extends generally in the first direction transverse to the first fuselage frame.

5. The securement assembly of claim 3, wherein the second end portion of the first clevis member comprises a first securement flange which extends along a forward side of the first fuselage frame with the first fastener extending through the first securement flange and the first fuselage frame.

6. The securement assembly of claim 2, wherein the first clevis member includes a first clevis portion and a second clevis portion, wherein the first opening of the first clevis member is further defined as a plurality of first openings.

7. The securement assembly of claim 6, wherein the first pin includes a plurality of first pins.

8. The securement assembly of claim 6, wherein: the second end portion of the first clevis portion comprises a first securement flange and the second end portion of the second clevis portion comprises a second securement flange; the first securement flange extends along a forward side of the first fuselage frame; and the second securement flange extends along the forward side of the first fuselage frame.

9. The securement assembly of claim 2, wherein: the second opening of the second clevis member is further defined as a plurality of second openings; the second clevis member is spaced apart from the first clevis member along a length of the aircraft; the second clevis member has a first prong which defines one of the plurality of second openings at the first end portion of the second clevis member and has a second prong which defines another one of the plurality of second openings at the first end portion of the second clevis member; and the second lug member is positioned between the first prong and the second prong of the second clevis member.

10. The securement assembly of claim 9, wherein the second pin extends through the second opening of the first prong of the second clevis member, the second opening of the second prong of the second clevis member and through a second lug opening defined by and through the second lug member engaging the second clevis member to the second lug member such that the second pin extends in a different direction than the first pin.

11. The securement assembly of claim 9, wherein the second end portion of the second clevis member comprises a third securement flange which extends along a forward side of the second fuselage frame with the second fastener extending through the third securement flange and the second fuselage frame in a direction transverse to the second fuselage frame.

12. The securement assembly of claim 9, wherein the second clevis member includes a third clevis portion and a fourth clevis portion secured to the second fuselage frame positioned spaced apart from one another in a third direction along the second fuselage frame.

13. The securement assembly of claim 12, wherein: the third clevis portion has a first prong at a first end portion of the third clevis portion; the third clevis portion has a second prong at the first end portion of the third clevis portion; the fourth clevis portion has a first prong at a first end portion of the fourth clevis portion; and the fourth clevis portion has a second prong at the first end portion of the fourth clevis portion.

14. The securement assembly of claim 13, wherein the second pin includes a plurality of second pins.

15. The securement assembly of claim 13, wherein: a second end portion of the third clevis portion comprises a third securement flange and a second end portion of the fourth clevis portion comprises a fourth securement flange.

16. A method for securing a vertical tail fin assembly to an aircraft, comprising: securing a first lug member, which is secured to the vertical tail fin assembly, to a first end portion of a first clevis member; and securing a second end portion of the first clevis member to a first fuselage frame, wherein the first fuselage frame is constructed of a composite material, wherein the second end portion of the first clevis member overlaps part of the first fuselage frame in which a first fastener extends through the second end portion of the first clevis member and the first fuselage frame in a first direction transverse to the first fuselage frame, wherein the composite material comprises multiple groups of a five-ply configuration which includes: one ply having a nonlinear fiber configuration; two plies having a first linear fiber configuration which extends within an angular range which includes plus or minus five degrees of plus thirty degrees of angular displacement from a radial axis of the nonlinear fiber configuration; and another two plies having a second linear fiber configuration which extends within an angular range which includes plus or minus five degrees of minus thirty degrees of angular displacement from the radial axis of the nonlinear fiber configuration.

17. The method for securing of claim 16, wherein the first lug member is secured to the first end portion of the first clevis member via a first pin, wherein the first end portion of the first clevis member defines a first opening along a first clevis axis in which the first pin extends through the first opening of the first clevis member and the first lug member to secure together the first clevis member and the first lug member, wherein the method further comprises: securing a second lug member, which is secured to the vertical tail fin assembly, to a first end portion of a second clevis member via a second pin, wherein the first end portion of the second clevis member defines a second opening along a second clevis axis in which the second pin extends through the second opening of the second clevis member and the second lug member to secure together the second clevis member and the second lug member, wherein the first clevis axis and the second clevis axis are spaced apart; securing a second end portion of the second clevis member to a second fuselage frame, wherein the second fuselage frame is constructed of the composite material, wherein the second end portion of the second clevis member overlaps part of the second fuselage frame in which a second fastener extends through the second end portion of the second clevis member and the second fuselage frame in the first direction transverse to the second fuselage frame; securing a third lug member, which is secured to the vertical tail fin assembly, to a first end portion of a third clevis member via a third pin, wherein the first end portion of the third clevis member defines a third opening along a third clevis axis in which the third pin extends through the third opening of the third clevis member and the third lug member to secure together the third clevis member and the third lug member, wherein the second clevis axis and the third clevis axis are spaced apart and substantially parallel to each other; securing a second end portion of the third clevis member to a third fuselage frame, wherein the third fuselage frame is constructed of the composite material, wherein the second end portion of the third clevis member overlaps part of the third fuselage frame

in which a third fastener extends through the second end portion of the third clevis member and the third fuselage frame in the first direction transverse to the third fuselage frame; securing a fourth lug member, which is secured to the vertical tail fin assembly, to a first end portion of a fourth clevis member via a fourth pin, wherein the first end portion of the fourth clevis member defines a fourth opening along a fourth clevis axis in which the fourth pin extends through the fourth opening of the fourth clevis member and the fourth lug member to secure together the fourth clevis member and the fourth lug member, wherein the first clevis axis and the fourth clevis axis spaced apart and substantially parallel to each other; and securing a second end portion of the fourth clevis member to a fourth fuselage frame, wherein the fourth fuselage frame is constructed of the composite material, wherein the second end portion of the fourth clevis member overlaps part of the fourth fuselage frame in which a fourth fastener extends through the second end portion of the fourth clevis member and the fourth fuselage frame in the first direction transverse to the fourth fuselage frame.

18. A securement assembly for securing a vertical tail fin assembly to an aircraft, comprising: a first lug member secured to the vertical tail fin assembly; and a first clevis member, wherein: a first end portion of the first clevis member is engaged to the first lug member; and a second end portion of the first clevis member is secured to a first fuselage frame, wherein the first fuselage frame is constructed of a composite material, wherein the composite material comprises: one ply having a nonlinear fiber configuration; two plies having a first linear fiber configuration which extends within an angular range which includes plus or minus five degrees of plus thirty degrees of angular displacement from a radial axis of the nonlinear fiber configuration; and another two plies having a second linear fiber configuration which extends within an angular range which includes plus or minus five degrees of minus thirty degrees of angular displacement from the radial axis of the nonlinear fiber configuration.

19. The securement assembly of claim 18, wherein the first end portion of the first clevis member defines a first opening along a first clevis axis in which a first pin extends through the first opening of the first clevis member and the first lug member to secure together the first clevis member and the first lug member, wherein the securement assembly further comprises: a second lug member secured to the vertical tail fin assembly; a second clevis member, wherein: a first end portion of the second clevis member is engaged to the second lug member, wherein the first end portion of the second clevis member defines a second opening along a second clevis axis in which a second pin extends through the second opening of the second clevis member and the second lug member to secure together the second clevis member and the second lug member; and a second end portion of the second clevis member is secured to a second fuselage frame, wherein the second end portion of the second clevis member overlaps part of the second fuselage frame in which a second fastener extends through the second end portion of the second clevis member and the second fuselage frame in a first direction transverse to the second fuselage frame; a third lug member secured to the vertical tail fin assembly; a third clevis member, wherein: a first end portion of the third clevis member is engaged to the third lug member, wherein the first end portion of the third clevis member defines a third opening along a third clevis axis in which a third pin extends through the third opening of the third clevis member and the third lug member to secure together the third clevis member and the third lug member; and a second end portion of the third clevis member is secured to a third fuselage frame, wherein the second end portion of the third clevis member overlaps part of the third fuselage frame in which a third fastener extends through the second end portion of the third clevis member and the third fuselage frame in the first direction transverse to the third fuselage frame; a fourth lug member secured to the vertical tail fin assembly; a fourth clevis member, wherein: a first end portion of the fourth clevis member is engaged to the fourth lug member, wherein the first end portion of the fourth clevis member defines a fourth opening along a fourth clevis axis in which a fourth pin extends through the fourth opening of the fourth clevis member and the fourth lug member to secure together the fourth clevis member and the fourth lug

member; and a second end portion of the fourth clevis member is secured to a fourth fuselage frame, wherein the second end portion of the fourth clevis member overlaps part of the fourth fuselage frame in which a fourth fastener extends through the second end portion of the fourth clevis member and the fourth fuselage frame in the first direction transverse to the fourth fuselage frame, and wherein the first pin and the fourth pin are spaced apart and orientated in the same direction relative to each other; and the second pin and the third pin are spaced apart and orientated in the same direction relative to each other.

20. The securement assembly of claim 19, wherein the first clevis axis and the fourth clevis axis are concentric to each other.
